

Perinatal Health Surveillance – United States, 2022

CDC Natality Data – Smoking, BMI, and Preterm Birth

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Contents

0.1	Data Source and Methods	1
0.2	Results	1
0.3	Conclusion	2

0.1 Data Source and Methods

The data are from the **CDC Natality Public Use File** for 2022, accessed via the NBER. We sampled 500,000 live births (stratified by preterm status) and stored them in a PostgreSQL star schema.

Definitions:

- **Preterm birth:** gestational age < 37 weeks (`gestrec3 == 1`).
- **Smoking pattern:** derived from cigarettes before pregnancy and during each trimester.
- **BMI category:** pre-pregnancy BMI (WHO cutoffs).
- **Risk ratios:** estimated using modified Poisson regression with robust standard errors.

0.2 Results

0.2.1 Table 1 – Maternal and Pregnancy Characteristics by Preterm Status

0.2.2 Prevalence of Preterm Birth by Smoking Pattern

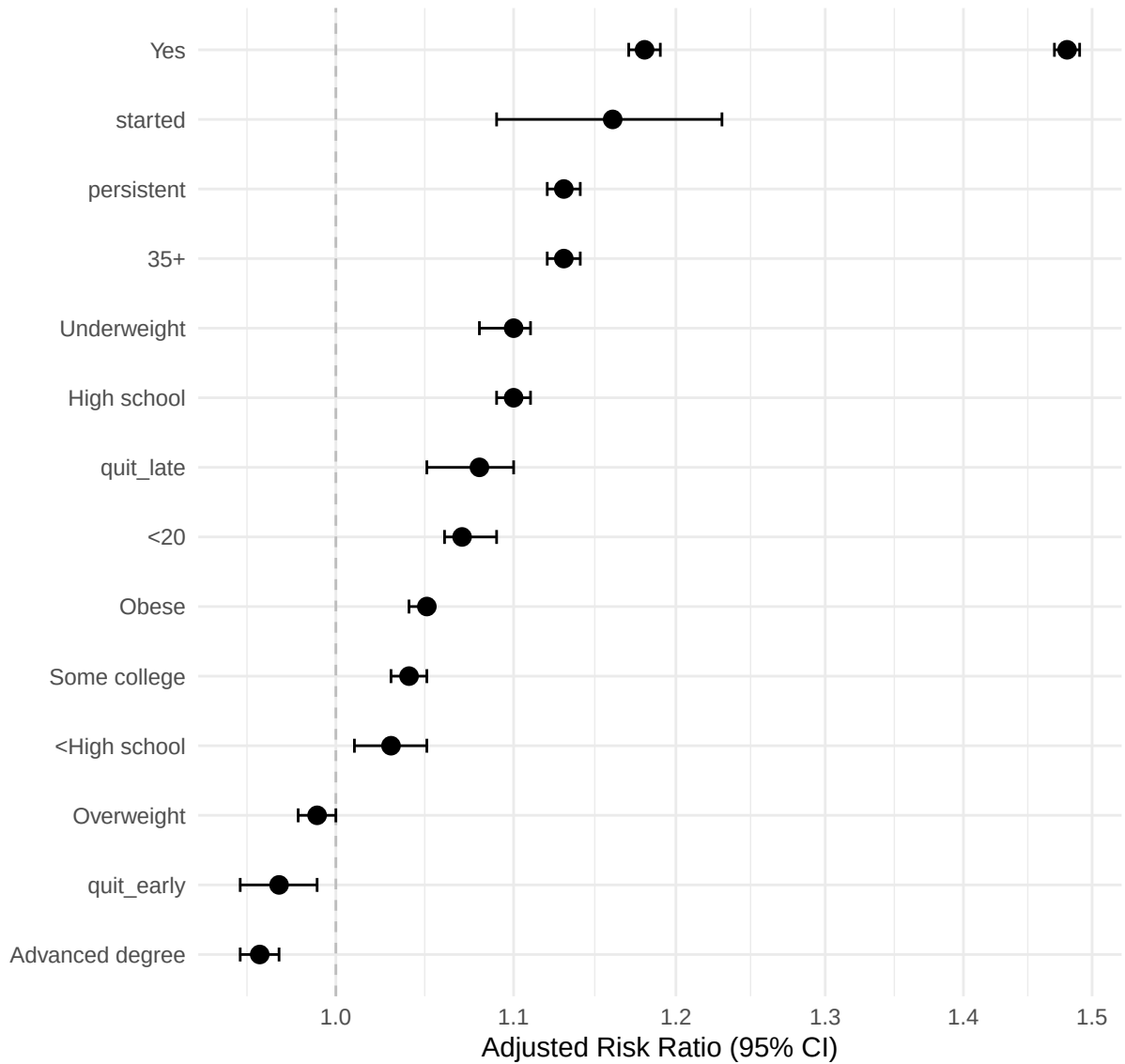
Table 1: Preterm birth prevalence by smoking pattern

smoking_pattern	n	preterm_n	preterm_pct
never	442814	214400	48.4
quit_early	9857	4876	49.5
quit_late	6861	3787	55.2
started	685	417	60.9
persistent	28965	17204	59.4

0.2.3 Risk Ratio Analysis

We model preterm birth (preterm vs. term) using modified Poisson regression with robust standard errors to obtain risk ratios.

0.2.4 Adjusted Risk Ratios (Forest Plot)



0.3 Conclusion

Persistent smoking during pregnancy is strongly associated with preterm birth, even after adjusting for BMI and other confounders. Obesity also increases the risk. This automated report can be regenerated whenever the underlying data update, providing a reproducible BI pipeline.

Characteristic	term N = 250,000 ^I	preterm N = 250,000 ^I	p-value
maternal_age_group			
20-34	193,270 (51%)	182,478 (49%)	
<20	11,719 (45%)	14,197 (55%)	
35+	45,011 (46%)	53,325 (54%)	
race_eth			
Hispanic	1,841 (44%)	2,389 (56%)	
Non-Hispanic Black	34,107 (39%)	53,154 (61%)	
Non-Hispanic Other	16,070 (55%)	13,319 (45%)	
Non-Hispanic White	130,810 (53%)	113,961 (47%)	
Unknown	67,172 (50%)	67,177 (50%)	
education			
College grad	49,029 (49%)	50,986 (51%)	
Some college	62,774 (47%)	71,020 (53%)	
High school	22,829 (43%)	30,322 (57%)	
<High school	7,664 (47%)	8,684 (53%)	
Advanced degree	20,833 (51%)	20,146 (49%)	
Unknown	86,871	68,842	
bmi_cat			
Normal	104,066 (53%)	92,942 (47%)	
Underweight	7,662 (47%)	8,750 (53%)	
Overweight	65,617 (51%)	63,110 (49%)	
Obese	72,655 (46%)	85,198 (54%)	
prior_preterm	6,944 (24%)	22,156 (76%)	
diabetes	17,971 (40%)	27,194 (60%)	
c_section			
Vaginal	175,422 (57%)	130,279 (43%)	
C-section	74,578 (38%)	119,721 (62%)	
sex			
Male	127,069 (49%)	132,510 (51%)	
Female	122,931 (51%)	117,490 (49%)	

^In (%)